

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - V

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Class :SYIT	Batch : 01
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Binary Search Tree: Write a program to:

- a. Create a binary search tree.

Code:

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')

root.left = nodeA
root.right = nodeB

nodeA.left = nodeC
nodeA.right = nodeD

nodeB.left = nodeE
nodeB.right = nodeF

nodeF.left = nodeG

print("root.right.left.data:", root.right.left.data)
print("NirajRane_S032")
```

Output:

```
> == RESTART: C:/Users/mvlui/Python37/Python37-64/Python.exe
root.right.left.data: E
NirajRane_So32
```

- b. Insert nodes into a binary search tree.

Code:

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')
nodeR = TreeNode('R')

root.left = nodeA
root.right = nodeB

nodeA.left = nodeC
nodeA.right = nodeD

nodeB.left = nodeE
nodeB.right = nodeF

nodeF.left = nodeG
nodeF.right = nodeR
print("root.right.right.right.data:", root.right.right.right.data)
print("NirajRane_So32")
```

Output:

```
> == RESTART: C:/Users/mvlui/Python37/Python37-64/Python.exe
root.right.right.right.data: R
NirajRane_So32
```

- c. Search for a node in a binary search tree.

Code:

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')

root.left = nodeA
root.right = nodeB

nodeA.left = nodeC
nodeA.right = nodeD

nodeB.left = nodeE
nodeB.right = nodeF

nodeF.left = nodeG

print("root.right.left.data:", root.right.left.data)

def search(root, key):
    if root is None:
        return False
```

```
    print("Checking:", root.data)

    if root.data == key:
        return True

    return search(root.left, key) or search(root.right, key)

key = input("Enter element to search: ")

if search(root, key):
    print("Element Found")
else:
    print("Element Not Found")

print("NirajRane_S032")
```

Output:

```
== RESTART: C:/Users/mvlui/One  
root.right.left.data: E  
Enter element to search: G  
Checking: R  
Checking: A  
Checking: C  
Checking: D  
Checking: B  
Checking: E  
Checking: F  
Checking: G  
Element Found  
NirajRane_S032
```

```
== RESTART: C:/Users/mvlui/One  
root.right.left.data: E  
Enter element to search: S  
Checking: R  
Checking: A  
Checking: C  
Checking: D  
Checking: B  
Checking: E  
Checking: F  
Checking: G  
Element Not Found  
NirajRane_S032
```